

Local update rules find stuck points, not minima: free-boundary partitions, and a remedy

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Abstract

A class of solvers partitions space into domains, assigns a field to each, and evolves the interfaces by a *local* rule: a cell changes ownership when a neighbouring domain's field exceeds its own, or when a locally computed velocity points that way. Level-set schemes with local reinitialisation, Voronoi-partitioned solvers with density-balance interfaces, and free-boundary formulations of multi-domain energy minimisation all have this structure.

Such a rule cannot minimise. It descends to the nearest configuration that no single admissible change improves — a *stuck point* — and which one it reaches depends on the initial condition. This is reported here as a measured effect rather than a possibility. In a multi-domain Coulomb energy minimisation, four independent geometries give converged energies that differ by 0.005–0.05 between seeds, between initial partitions, and between binary builds differing only in floating-point rounding, against target quantities of 0.001–0.02 in the same units. The ambiguity exceeds the quantity sought by one to two orders in every case. Each run converges cleanly; they converge to different places.

Four remedies were tried and none works: an interfacial energy penalty regularising the boundary, a five-fold reduction of the interface time step, freezing the interface entirely, and changing to a physical system without the first one's pathologies. The reason they fail is that this is a defect of the *search* and not of the energy functional, so nothing expressed in the functional reaches it.

The remedy that does work is to stop searching. Interfaces in these problems are surfaces and can be placed: specify the partition, hold it fixed, relax only the fields, and evaluate what a stated configuration costs rather than asking where a boundary goes. Repeated runs then agree to every digit recorded, and quantities that were previously irreproducible become well defined — a localised defect energy independent of system size to 5%, and an energy of delocalisation independent of it likewise. The cost is that the family of admissible partitions must be constructed rather than discovered, and three constructions are given, together with the invariants that rejected two of them.

1 The structure at issue

Consider a domain of space divided into subdomains $\Omega_1, \dots, \Omega_N$ with $\bigcup \Omega_i = \Omega$ and $\Omega_i \cap \Omega_j = \emptyset$, a field u_i supported on each, and an energy $E[\{u_i\}, \{\Omega_i\}]$ to be minimised over both the fields and the partition. The fields are handled by any standard relaxation. The partition is the difficulty, and the usual treatment is an interface rule of the form

$$\text{cell } c \text{ moves from } \Omega_i \text{ to } \Omega_j \quad \text{when} \quad w_{ij}(c) < 0, \quad (1)$$

where w_{ij} is computed from the fields in a neighbourhood of c — a density balance, a local energy difference, a level-set velocity. The rule is attractive because it is local, cheap, and requires no global bookkeeping.

It is also, as an optimiser, a descent method whose moves are single-cell transfers. Its fixed points are the configurations from which no single such transfer decreases the energy. These are not the minima of E over partitions, and there is in general no relation between the two sets beyond inclusion of the latter in the former. That much is elementary. What is not obvious in advance is the *size* of the resulting ambiguity in a problem of practical interest, and whether it can be controlled by the means one would ordinarily reach for.

2 The measurement

The setting is a minimisation of the Coulomb energy of N non-overlapping unit charge densities, one per domain, with the interfaces evolved by (1) using a density balance $w = (\rho_i - \rho_j)/(\rho_i + \rho_j)$. Energies are in Hartree. Four geometries were run, each many times, varying only the initial condition or the compiled binary:

geometry	what was varied	spread	target quantity
ring of six sites	the binary, physics identical	0.005	0.012–0.022
chain of nine sites	— (never settles)	0.05	~ 0.01
two-centre transfer	four initial partitions	0.031	~ 0.001
chain of five sites	two initial partitions	0.013	~ 0.001

Three remarks on the entries. The ring’s spread appeared between two builds of the *same source* differing in an unrelated addition; recompilation perturbs floating-point rounding, cells sitting near the threshold in (1) then fall the other way, and a different stuck point follows. The chain of nine did not converge at all, oscillating between two partitions indefinitely, which is the same phenomenon with a two-cycle instead of a fixed point. The two-centre transfer was monotone in the initial displacement of one domain — four seeds, four energies, ordered — which is the signature of a landscape of stuck points rather than of noise.

Each individual run is deterministic and reproducible: repeating one with the same seed and the same binary returns the same energy to every digit. The irreproducibility is entirely across initial conditions, and it exceeds the quantities being computed by one to two orders in all four cases.

3 What does not fix it

An interfacial penalty. Adding $\frac{\beta}{2} \oint u^2 dS$ to the energy gives the interface a restoring force and is the natural regularisation. It behaves correctly as an energy: at fixed initial condition it raises the total, linearly in β at small β . It does not reduce the spread. In the two-centre geometry the spread across seeds went from 0.031 at $\beta = 0$ to 0.040 at $\beta = 0.05$ — slightly worse. Spreading is an energy question and the penalty answers it; which stuck point is reached is a search question, on which the penalty is silent.

A smaller interface step. Reducing the interface time step five-fold changes nothing. Oscillation between two partitions looks like an overshoot and is not: the two configurations are both fixed points of the fields, and the cycle is between them, not around one.

Freezing the interface. Holding the partition fixed at its initial value removes the search but also removes the answer, since the initial partition is then simply asserted. Applied to the oscillating chain it left a residual variation of the same order, which localised the remaining difficulty in the *field* relaxation for that geometry — a separate defect, and evidence that the interface rule is not the only source of trouble in multi-domain solvers.

A different physical system. The first geometries involved a weakly bound outer domain whose field could lower its energy by spreading without limit, and one might attribute the spread to that pathology. Replacing it with a system where every domain sits in a confining well removed the pathology — energies became physically sensible, with no runaway — and the seed spread remained, at 0.013. The effect is not a property of the particular system.

And geometry cannot bound it either. Confining domains to prescribed regions to prevent spreading fails structurally when the partition must cover the domain: excluding a region from every subdomain leaves it owned by whichever label the implementation defaults to. In our case that produced a constrained minimum *below* the unconstrained one — impossible, and a useful inequality to test against, since it exposed the error immediately.

4 The remedy: specify the partition

Interfaces in these problems are surfaces, and a surface can be placed. For a chain geometry the interfaces are planes transverse to the axis, so a partition is fixed by the plane positions; more generally any parametrised family will do. The procedure is:

1. construct a family of admissible partitions $\Omega(\lambda)$ parametrised by a coordinate λ carrying the physics of interest;
2. for each λ , fix the partition and relax the fields alone to convergence;
3. read $E(\lambda)$, and take from it whatever the problem requires — a minimum, a barrier as the amplitude over a period, a formation energy as a difference against a reference partition.

Nothing is searched for, so nothing can be stuck. The question changes from “where does the boundary go” to “what does this configuration cost”, which is the question a barrier or a formation energy actually poses. Repeated runs agree to every digit recorded — verified three times, at two system sizes, against the 0.005–0.05 spreads above.

What it then delivers. In the same multi-domain problem: a localised interface defect costing 0.0064 and 0.00675 at two system sizes, i.e. independent of size to 5% while the total energy changes by 0.26 between them — a factor of 750 in sensitivity; and an energy of delocalisation of 0.111 and 0.113 at those two sizes, agreeing to 2%. Neither quantity was obtainable at all by the free-boundary route, both being smaller than its ambiguity.

What it costs. The family $\Omega(\lambda)$ must be constructed, and constructing it correctly is not automatic. Three families were built for a single physical question and two were wrong. One left a subdomain stranded against the boundary of Ω with no source in it; one collapsed a subdomain to zero measure; the third was correct. Each was rejected before being run, by checking an invariant the family had to satisfy — that the ends of a chain be identical between the configurations

compared, that the subdomain count be preserved, that a translation by one period return the original energy. *Constructing the invariant is the substantive work*, and it is worth doing before the computation rather than after, since a family that violates one produces numbers that look entirely reasonable.

5 Scope

The finding applies to any scheme whose partition is advanced by a rule of the form (1), which includes local level-set reinitialisation, Voronoi- or power-diagram solvers with locally computed interface velocities, and multi-domain energy minimisations of the kind above. It does not apply to schemes that optimise the partition globally, nor to those in which the partition is determined by the physics rather than sought — a symmetry-fixed interface, for example, is exact and costs nothing.

The remedy is available whenever the partitions of interest form a family that can be written down. Where the partition itself is the unknown and no such family is available, the difficulty is real and this note offers no solution to it; the honest course there is to report the ambiguity rather than a converged energy, and to test for it, which requires only running the same problem twice from different initial conditions. That test is cheap, it is not standard practice, and in four geometries out of four here it would have been decisive.

6 Conclusion

A local interface rule is a descent method over single-cell moves, and its fixed points are stuck points rather than minima. In a multi-domain Coulomb minimisation the resulting ambiguity was 0.005–0.05 Hartree across four geometries, one to two orders above the quantities being computed, and was not removed by an interfacial penalty, a smaller step, a frozen interface, or a change of physical system — because it is a property of the search and not of the energy. Specifying the partition and relaxing only the fields restores reproducibility to every recorded digit and makes quantities computable that were not computable otherwise. The price is that the family of partitions must be constructed and checked against an invariant, and the check must precede the computation.